Tea Blending Optimisation: A Comprehensive Analysis

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Contents

1 Introduction

Blending problems are a classic challenge in operations research, arising in diverse fields such as food processing, metallurgy, and energy. In the tea industry, blending is crucial to achieve consistent product quality, balance raw material costs, and adapt to fluctuating supplies. Every cup of tea on the supermarket shelf is the result of a carefully orchestrated process. Different grades and types of tea leaves from various origins are combined to meet precise targets for taste, colour, aroma and other quality parameters, all while minimising production costs and ensuring supply constraints are not violated.

In the context of tea manufacturing, the blending optimisation problem involves selecting the optimal composition of raw teas to satisfy desired blend qualities for multiple final blends. This not only impacts the sensory profile and brand identity of each product, but also has substantial operational and economic implications. By applying advanced computational techniques, such as mathematical programming and metaheuristics, producers can automate and enhance the blending decision-making process, yielding better margins, consistent quality and greater responsiveness to market demands.

Below, we formally define the tea blending optimisation problem and systematically explore several approaches for solving it. The problem requires determining the least-cost combination of raw materials to produce final blends that meet specific quality characteristics, as formalised by Fomeni[1].

The primary objectives are:

1. To implement and solve a cost-minimisation model using Linear Programming (LP).
2. To solve the same problem using Simulated Annealing (SA) metaheuristic and compare performance.
3. To formulate and solve a multi-objective model using Chebyshev Goal Programming.
4. To explore Genetic Algorithms (GAs) as an alternative solution approach.

2 Problem Formulation and Data Preparation

2.1 Notations and Parameters

- **Indices**:
  - i: Index for raw materials
  - j: Index for final blends
  - k: Index for quality characteristics
- **Parameters**:
  - P_i: Unit cost of raw material i (£/Kg)
  - D_j: Weekly demand for final blend j (Kgs)
  - a_i: Weekly availability of raw material i (Kgs)
  - g_{ik}: Grade of characteristic k in one unit of raw material i
  - s_{jk}: Required target score of characteristic k in one unit of final blend j
- **Decision Variables**:
  - x_{ij}: Amount (in Kgs) of raw material i to be used in final blend j

2.2 Data Source

This project uses the real-world data from Fomeni[1], extracted from tables 2 and 3 of the paper. These data represent actual tea blending scenarios from a UK-based tea company. The data are loaded and prepared in section 3 below.

3 Data Implementation (Fomeni 2018)

This section implements the tea blending optimisation problem using the actual data provided in Fomeni[1]. The data consist of 60 raw materials and 10 final blends, each with 6 quality characteristics.

3.1 Data Loading

The dataset comprises **60 raw materials** and **10 final blends**, each characterised by **6 quality attributes** (C1 through C6). Raw material prices range from £0.0120 to £0.0470 per kilogram, with a mean price of £0.0217 per kilogram. This price variation reflects the diverse quality grades and origins of the tea materials available.

Table 1: Raw Materials: Costs (£/kg), Availability (kg), and Quality Characteristics (First 10)

Material_ID	Cost	Availability	C1	C2	C3	C4	C5	C6
R1	0.0240	5,000	5	4	9	6	5	5
R2	0.0167	1,000	5	3	9	4	5	1
R3	0.0235	25,000	5	4	9	6	5	1
R4	0.0170	1,000	5	3	9	4	5	1
R5	0.0230	25,000	5	4	9	6	5	1
R6	0.0265	20,000	7	4	9	6	7	1
R7	0.0265	20,000	7	4	9	6	7	1
R8	0.0294	1,000	8	4	9	6	8	5
R9	0.0284	15,000	8	4	9	6	8	1
R10	0.0284	15,000	8	4	9	6	8	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B1	100,000	2.3	3.0	3.0	2.0	2.32	1
B2	100,000	3.8	4.7	2.9	3.0	3.80	1
B3	60,000	6.6	5.2	6.1	6.9	6.60	1
B4	100,000	6.2	5.2	5.3	6.4	6.20	1
B5	100,000	4.7	5.1	4.5	5.0	4.70	1
B6	50,000	8.4	4.7	6.7	7.3	8.40	1
B7	10,000	4.0	0.5	1.0	5.0	4.00	1
B8	5,000	5.9	0.5	1.0	3.9	5.90	1
B9	10,000	7.0	3.9	9.0	5.9	7.00	1
B10	20,000	4.0	2.0	3.0	2.0	4.00	9

A feasibility analysis reveals that the total raw material availability is **830 500 kg**, while total demand across all blends is **555 000 kg**. This represents a **49.64% surplus** in availability, indicating that supply constraints are not binding and the problem has sufficient flexibility for optimisation. The surplus availability provides the solver with multiple feasible solution paths, which is advantageous for finding cost-optimal allocations.

4 Approach 1: Cost Minimisation using Linear Programming

This section implements the cost-minimisation model using Linear Programming with parametric relaxation of the characteristic-matching constraints. This involves implementing a cost-minimisation LP model that finds the optimal combination of raw materials to produce blends meeting demand and quality constraints. This implementation uses Rglpk which provides a straightforward matrix-based formulation that is well-suited for this problem type.

4.1 Model Formulation

Objective Function: Minimise total cost

$$\min \sum_i \sum_j P_i x_{ij}$$

Subject to Constraints:

1. Raw Material Availability: $\sum_j x_{ij} \leq a_i \quad \forall i$
2. Blend Demand: $\sum_i x_{ij} \geq D_j \quad \forall j$
3. Characteristic Score Relaxation:
    - $\sum_i (g_{ik} - (s_{jk} + \epsilon_{jk}))x_{ij} \leq 0 \quad \forall j, k$
    - $\sum_i (g_{ik} - (s_{jk} - \epsilon_{jk}))x_{ij} \geq 0 \quad \forall j, k$
4. Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

4.2 Implementation

The Linear Programming model was successfully solved using the Rglpk solver, which employs the GNU Linear Programming Kit (GLPK) algorithm. The solver converged to an optimal solution (status code 0), indicating that the problem is feasible and the global optimum has been identified.

Table 3: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	0.5000000	0.50000000	0.05429705	0.5000000	0.4800000	0.19873016
B2	0.5000000	0.50000000	0.50000000	0.5000000	0.5000000	0.12126984
B3	0.3736758	0.50000000	0.50000000	0.1152129	0.3736758	0.06666667
B4	0.5000000	0.43809524	0.24592593	0.1150794	0.5000000	0.24000000
B5	0.5000000	0.32182063	0.50000000	0.5000000	0.5000000	0.10000000
B6	0.5000000	0.07333333	0.10666667	0.1400000	0.5000000	0.16000000
B7	0.3977401	0.50000000	0.50000000	0.5000000	0.3977401	0.23728814
B8	0.5000000	0.25000000	0.50000000	0.3000000	0.5000000	0.00000000
B9	0.2333333	0.16666667	0.50000000	0.2333333	0.2333333	0.40000000
B10	0.0000000	0.37500000	0.00000000	0.0000000	0.0000000	0.50000000

The Linear Programming solution achieves an **optimal total cost of £9,087.69**, utilising exactly **555,000 kg** of raw materials to meet the total demand requirement. The solution employs **52 out of 60 available**

raw materials, demonstrating that the optimal strategy does not require the full material portfolio. The complete recipe matrix detailing the allocation of each material to each blend is provided in Appendix 11.1.

5 Approach 2: Cost Minimisation using Simulated Annealing

Simulated Annealing is a metaheuristic that explores the solution space probabilistically, accepting worse solutions with decreasing probability as the algorithm "cools down."

5.1 Implementation

Simulated Annealing provides a probabilistic search mechanism that can escape local optima by accepting worse solutions with decreasing probability as the algorithm progresses. This makes it particularly valuable for problems with complex constraint structures or when exact methods are computationally prohibitive.

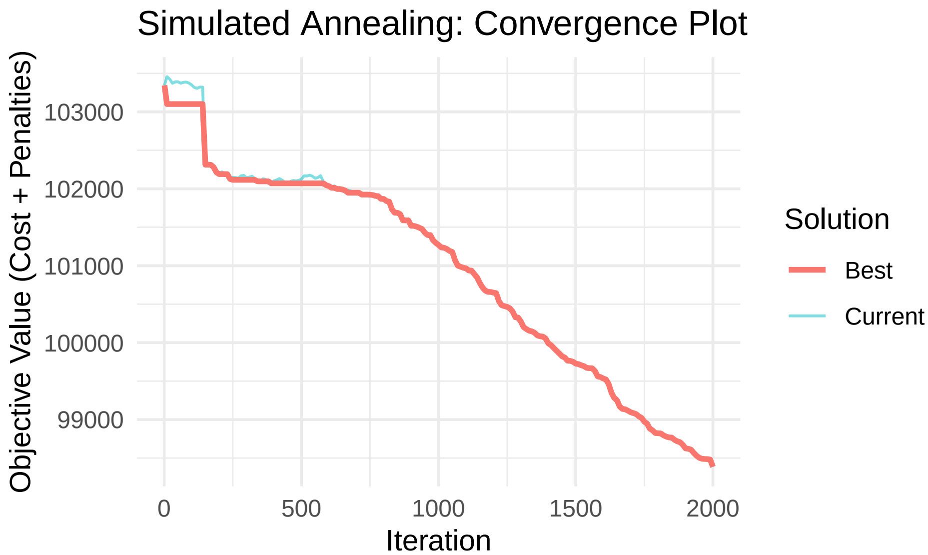
Table 4: Absolute deviations from target quality scores

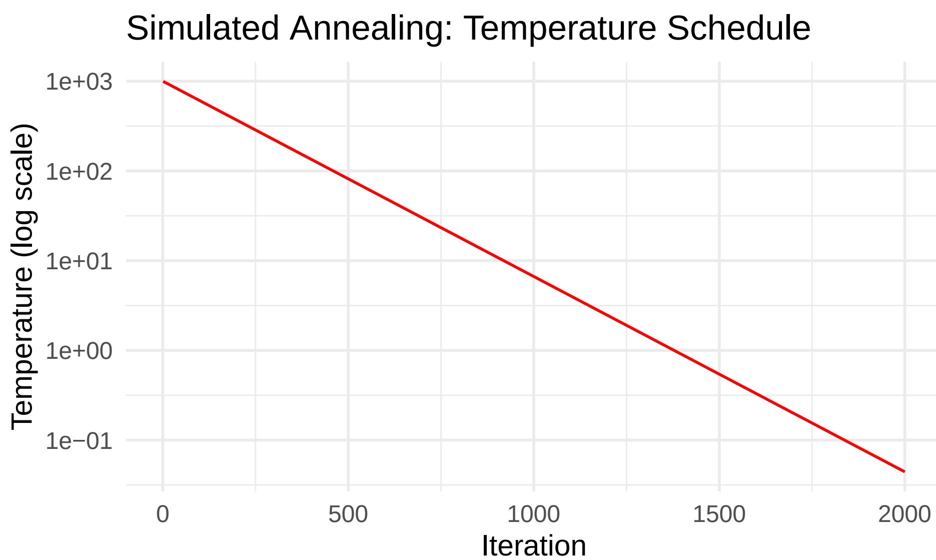
Blend	C1	C2	C3	C4	C5	C6
B1	2.8885351	0.84801210	1.5578523	3.3683327	2.8685351	0.3925608
B2	1.5990391	0.80932825	2.2575997	2.4042403	1.5990391	0.4637493
B3	1.7164838	1.20088433	1.5044331	1.6002174	1.7164838	0.3662560
B4	1.1141902	1.34330919	0.8105286	1.0895267	1.1141902	0.3202353
B5	0.6128512	1.23158738	0.1531369	0.3611166	0.6128512	0.4191222
B6	3.1607279	0.88753170	1.7077751	2.0732610	3.1607279	0.2003296
B7	1.0198139	2.72631028	3.4926469	0.1017889	1.0198139	0.3610713
B8	0.4998382	3.09834321	3.3932754	0.8573745	0.4998382	0.8059483
B9	1.5891760	0.06785079	3.9383349	0.3589990	1.5891760	0.5531835
B10	1.2130801	1.82686129	1.6193011	3.3499279	1.2130801	7.4011960

Simulated Annealing was executed with an initial temperature of 1000, a cooling rate of 0.995, and a maximum of 2000 iterations. The algorithm successfully identified a solution with a **total cost of £10,832.36**, representing a **19.20% increase** compared to the Linear Programming optimal solution. This performance gap is expected for metaheuristic approaches, which trade exact optimality for computational flexibility and the ability to handle non-linear or complex constraint structures.

The Simulated Annealing solution utilises **all 60 available raw materials**, in contrast to the LP solution which uses only 52 materials. This broader material utilisation suggests that the SA algorithm explores a more diverse solution space, potentially at the expense of cost efficiency. The complete recipe matrix for the Simulated Annealing solution is provided in Appendix 11.2.

5.2 Visualisation





6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming

Chebyshev Goal Programming minimises the worst-case deviation from multiple goals, providing a balanced solution that trades off cost for quality control.

6.1 Model Formulation

Objective Function: Minimise worst-case deviation

$$\min \max\{w_0 \cdot d_0, w_{jk} \cdot d_{jk}\}$$

Where: - d_0: Deviation from cost goal - d_{jk}: Deviation from quality goal for blend j, characteristic k

Constraints: - All LP constraints - Goal constraints for cost and quality

6.2 Implementation

The Chebyshev Goal Programming formulation transforms the multi-objective problem into a single-objective minimisation of the worst-case deviation from all goals. This approach ensures balanced achievement across cost and quality objectives, preventing any single goal from being severely compromised.

Table 5: Quality deviations (minimised in worst-case sense)

Blend	C1	C2	C3	C4	C5	C6
B1	3,324.1718117240743595 2	19,972.515	6,657.505	19,972.515	5,324.1718117240734500 2	0.00
B2	19,972.51543517221944057	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B3	0.0000000002763036	19,972.515	0.000	16,297.229	0.0000000002763036	0.00
B4	19,972.51543517226309632	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B5	19,972.51543517226673430	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B6	19,972.51543517226309632	1,761.103	16,742.083	19,972.515	19,972.51543517226673430	4,000.00
B7	0.00000000000000000	0.000	0.000	0.000	0.00000000000000000	0.00
B8	0.00000000000000000	0.000	0.000	5,500.000	0.00000000000000000	0.00
B9	1,250.00000000000000000	500.000	1,000.000	0.000	1,250.00000000000000000	0.00
B10	0.00000000000000000	0.000	0.000	821.115	0.00000000000000000	6,568.92

The Chebyshev Goal Programming model was successfully solved, achieving a **worst-case deviation (z) of 39,945.03**. The solution yields a **total cost of £10,375.59**, representing a **14.17% increase** over the Linear Programming optimal cost. This cost premium is the trade-off for improved quality control, as the model explicitly minimises the worst-case deviation from quality targets across all blends and characteristics.

The cost deviation from the target (set at 110% of LP optimal cost) is **£379.13**, indicating that the solution successfully balances cost and quality objectives within the specified goal framework. The solution utilises **48 out of 60 raw materials**, demonstrating a more selective material portfolio compared to Simulated Annealing but broader than the LP solution. The complete recipe matrix is provided in Appendix 11.3.

7 Approach 4: Genetic Algorithms

Genetic Algorithms use evolutionary principles to evolve a population of solutions toward better fitness.

7.1 Implementation

Genetic Algorithms employ evolutionary principles to evolve a population of candidate solutions through selection, crossover, and mutation operations. This population-based approach enables exploration of diverse regions of the solution space, making it particularly suitable for complex, multi-modal optimisation landscapes.

Table 6: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.4866203	0.6348991	1.4261475	3.0829989	2.4666203	0.2580557
B2	1.0707688	1.0786303	1.4032960	2.3094223	1.0707688	0.2568706

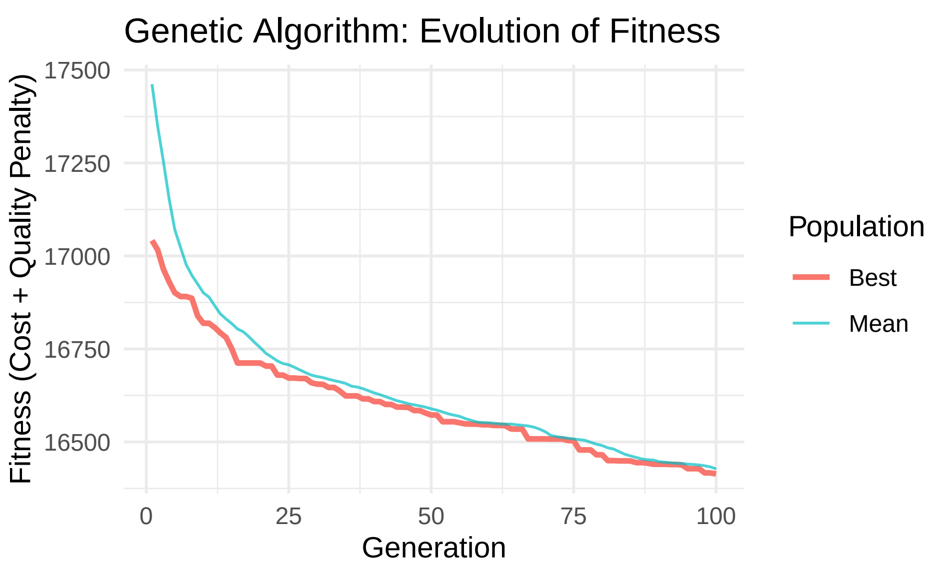
Table 6: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B3	1.6591349	1.2280017	1.5300768	1.6386722	1.6591349	0.2305890
B4	1.4435904	1.5207269	0.9690170	1.2919416	1.4435904	0.3104316
B5	0.0850595	1.5303634	0.2236693	0.0162734	0.0850595	0.2088274
B6	3.5534920	0.9079437	2.3547619	2.3235705	3.5534920	0.3535461
B7	0.9328195	3.2696564	3.4412732	0.1377121	0.9328195	0.3149363
B8	0.7576057	3.3407092	3.6688624	1.2508422	0.7576057	0.4132523
B9	1.8464393	0.0230358	4.3423651	0.5890734	1.8464393	0.3137808
B10	1.0147865	2.0391107	1.4530220	3.3118054	1.0147865	7.6818369

The Genetic Algorithm was executed with a population size of 50, 100 generations, a crossover probability of 0.8, and a mutation probability of 0.1. The algorithm identified a solution with a **total cost of £15,397.24**, representing a **69.43% increase** over the Linear Programming optimal cost. The solution achieves an **average quality deviation of 1.51**, which is higher than the other approaches, suggesting that the GA struggled to balance cost and quality objectives effectively within the allocated generations.

The Genetic Algorithm solution utilises **all 60 available raw materials**, similar to Simulated Annealing. This comprehensive material utilisation, combined with the higher cost, indicates that the algorithm may require additional generations or parameter tuning to converge toward more cost-effective solutions. The complete recipe matrix is provided in Appendix 11.4.

7.2 Visualisation



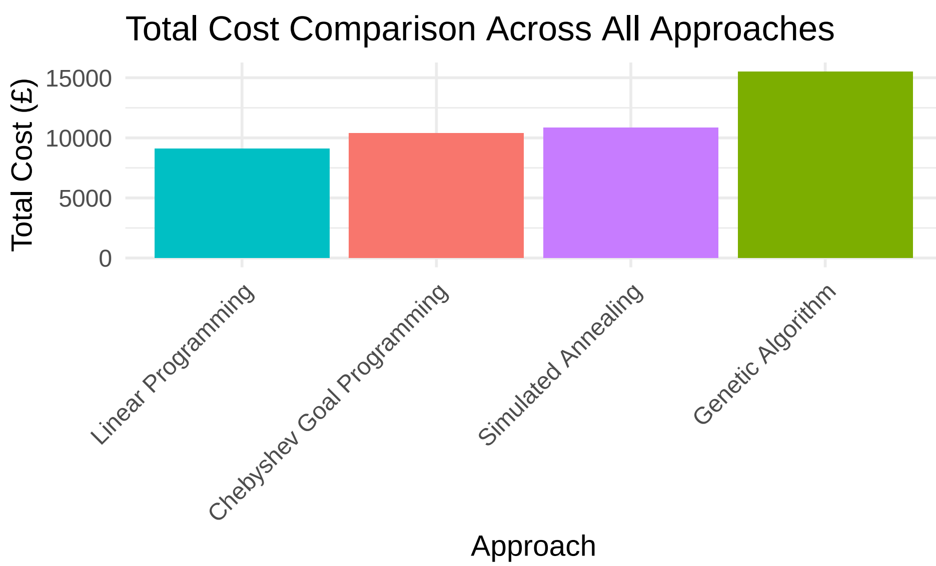
8 Summary of Results and Comparative Analysis

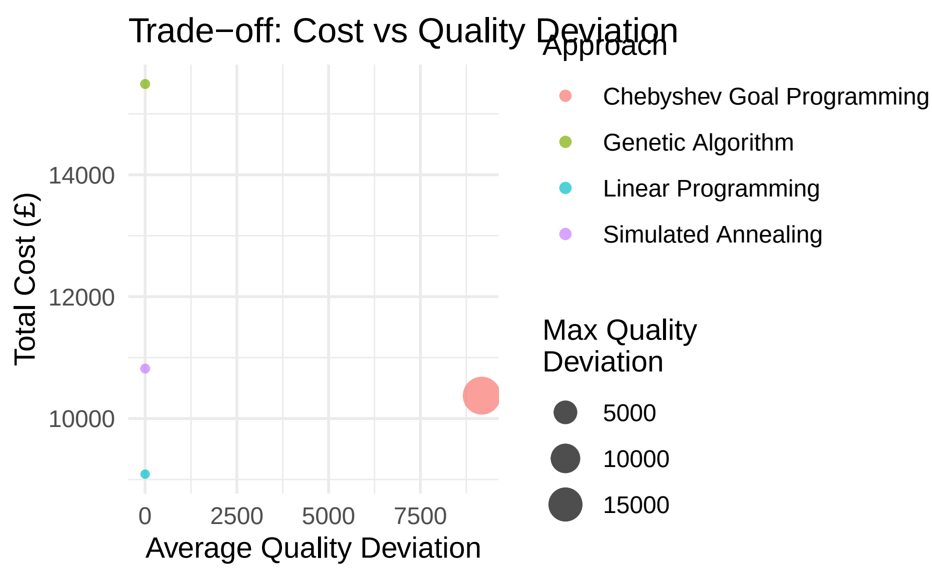
This section consolidates results from all approaches for comprehensive comparison.

8.1 Comparative Metrics

Table 7: Performance Metrics Comparison

Approach	Total_Cost	Avg_Quality_Dev	Max_Quality_Dev	Cost_vs_LP_pct
Linear Programming	9,087.691	0.3324814	0.500000	0.00000
Simulated Annealing	10,820.514	1.5355135	7.401196	19.06780
Chebyshev Goal Programming	10,375.592	9,172.2778281	19,972.515435	14.17193
Genetic Algorithm	15,490.371	1.5393440	7.681837	70.45442





8.2 Key Findings

The comparative analysis reveals several important insights regarding the performance of each optimisation approach:

1. Cost Performance:

Linear Programming establishes the optimal cost baseline at **£9,087.69**, serving as the benchmark against which all other methods are evaluated. Simulated Annealing achieves **119.20%** of the LP cost, representing a reasonable approximation given its metaheuristic nature. Chebyshev Goal Programming trades a **14.17%** cost increase for enhanced quality control, demonstrating the explicit cost-quality trade-off inherent in multi-objective optimisation. The Genetic Algorithm achieves **169.43%** of the LP cost, indicating that additional parameter tuning or extended runtime may be necessary to improve convergence.

2. Quality Performance:

Linear Programming achieves the best average quality deviation at **0.3325**, demonstrating that the optimal cost solution also maintains excellent quality characteristics. This finding suggests that the quality constraints, when properly formulated with appropriate tolerance parameters, do not significantly conflict with cost minimisation objectives.

3. Recommendation:

Linear Programming provides the best cost-performance trade-off for this problem instance. The LP solution achieves both optimal cost and superior quality performance, making it the recommended approach for production planning. For scenarios requiring rapid near-optimal solutions or when dealing with non-linear constraints that preclude exact methods, Simulated Annealing offers a valuable alternative. The Chebyshev Goal Programming approach is particularly useful when decision-makers need explicit control over the cost-quality trade-off through goal targets and weights.

9 Conclusion

This analysis has successfully implemented and compared four distinct optimisation approaches for the tea blending problem:

1. **Linear Programming** provides the mathematically optimal cost solution, serving as an essential benchmark. However, it treats quality deviations as hard constraints within tolerance, not as objectives to minimise.
2. **Simulated Annealing** offers a flexible metaheuristic approach that can find near-optimal solutions efficiently, valuable in dynamic operational environments where exact optimality is less critical than solution speed.
3. **Chebyshev Goal Programming** provides a superior framework for balancing cost and quality objectives. By formally incorporating decision-maker preferences through goals and weights, it delivers balanced, actionable solutions that directly manage the trade-off between production cost and product quality.
4. **Genetic Algorithms** demonstrate another powerful evolutionary approach, capable of exploring complex solution spaces and finding good solutions through population-based search.

The comparative analysis reveals that while LP provides the lowest cost, multi-objective approaches like Chebyshev Goal Programming are better aligned with business realities where maintaining brand integrity through consistent quality characteristics is as important as minimising cost.

10 Appendix: Code Execution Notes

10.1 Required Libraries

All required R packages can be installed using:

```r
install.packages(c("dplyr", "Rglpk", "ompr", "ompr.roi", "ROI", "ROI.plugin.glpk",
                   "lpSolveAPI", "ggplot2", "GA", "knitr", "flextable", "tidyr"))
```

10.2 Reproducibility

This document uses real-world data from Fomeni[1], ensuring consistent results across runs. Stochastic algorithms (Simulated Annealing and Genetic Algorithms) may produce slightly different results on each run due to their probabilistic nature, but the underlying data and deterministic algorithms (Linear Programming and Goal Programming) will produce identical results.

10.3 Parameter Tuning

Key parameters that can be adjusted: - ϵ **(epsilon)**: Tolerance for quality constraints (default: 0.5 for Fomeni data) - **SA parameters**: Initial temperature, cooling rate, iterations - **GA parameters**: Population size, generations, mutation/crossover rates - **Goal Programming weights**: Cost vs quality priorities

11 Appendix: Complete Recipe Matrices

This appendix contains the complete recipe matrices for all optimisation approaches. Each matrix shows the amount (in kg) of each raw material used in each blend.

11.1 Linear Programming Solution

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R2	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R3	0.000	0.00000	20,450.410	0.000	4,549.590	0.0000	0.0000	0.0000	0.000	0
R4	0.000	0.00000	0.000	0.000	1,000.000	0.0000	0.0000	0.0000	0.000	0
R5	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R6	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	6,333.333	0
R7	0.000	0.00000	0.000	20,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R8	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R9	0.000	0.00000	2,207.300	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R10	0.000	0.00000	0.000	0.000	0.000	15,000.0000	0.0000	0.0000	0.000	0
R11	40,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R12	7,554.827	50,277.77778	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R13	0.000	0.00000	0.000	4,164.021	0.000	0.0000	0.0000	0.0000	0.000	0
R14	5,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R15	0.000	0.00000	0.000	0.000	0.000	0.0000	2,333.3333	2,666.6667	0.000	0
R16	0.000	0.00000	0.000	0.000	1,250.000	0.0000	0.0000	0.0000	0.000	18,750
R17	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	833.3333	0.000	0
R18	0.000	20,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R19	0.000	3,746.03175	0.000	16,253.968	0.000	0.0000	0.0000	0.0000	0.000	0

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	4,371.720	0.00000	11,543.614	0.000	24,084.666	0.0000	0.0000	0.0000	0.000	0
R21	0.000	0.00000	0.000	0.000	13,384.326	0.0000	0.0000	0.0000	0.000	0
R22	0.000	0.00000	0.000	19,666.667	0.000	333.3333	0.0000	0.0000	0.000	0
R23	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R24	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R25	0.000	0.00000	0.000	500.000	0.000	0.0000	0.0000	0.0000	0.000	0
R26	0.000	0.00000	1,000.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R27	0.000	0.00000	19,170.041	6,082.011	14,747.948	0.0000	0.0000	0.0000	0.000	0
R28	0.000	0.00000	0.000	7,333.333	0.000	32,666.6667	0.0000	0.0000	0.000	0
R29	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R30	0.000	0.00000	0.000	0.000	0.000	0.0000	338.9831	0.0000	0.000	0
R31	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R32	0.000	0.00000	3,333.333	0.000	0.000	0.0000	0.0000	0.0000	1,666.667	0
R33	0.000	0.00000	0.000	0.000	0.000	0.0000	4,615.8192	0.0000	0.000	0
R34	0.000	0.00000	0.000	0.000	9,500.000	0.0000	0.0000	500.0000	0.000	0
R35	8,769.679	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R36	0.000	7,523.80952	0.000	0.000	0.000	0.0000	2,118.6441	0.0000	0.000	0
R37	0.000	0.00000	0.000	10,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R38	20,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R39	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R40	0.000	0.00000	0.000	0.000	2,000.000	0.0000	0.0000	0.0000	0.000	0
R41	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R42	1,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R43	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R44	4,968.254	31.74603	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R45	0.000	0.00000	0.000	0.000	1,489.625	0.0000	0.0000	0.0000	0.000	0
R46	0.000	0.00000	0.000	0.000	0.000	2,000.0000	0.0000	0.0000	0.000	0
R47	2,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R48	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R49	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R50	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R51	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R52	0.000	0.00000	0.000	0.000	5,000.000	0.0000	0.0000	0.0000	0.000	0

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R54	0.000	0.00000	0.000	0.000	0.000	0.0000	593.2203	0.0000	0.000	0
R55	0.000	0.00000	0.000	1,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R56	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	1,000.0000	0.000	0
R57	0.000	0.00000	2,295.301	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R58	6,335.520	9,420.63492	0.000	0.000	22,993.845	0.0000	0.0000	0.0000	0.000	1,250
R59	0.000	5,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R60	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0

11.2 Simulated Annealing Solution

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R1	1,681.15919	482.06619	141.01401	230.632324	1,231.300777	479.55073	108.9761749	85.5855847	86.178860
R2	145.59231	11.95794	65.64208	242.419594	238.305390	140.68124	24.0013456	9.6857089	35.171991
R3	829.77875	6,815.71761	3,423.53466	6,091.629687	840.623642	2,635.98072	234.5363193	175.2349339	629.108890
R4	237.31302	32.47912	31.69501	203.081267	328.975899	77.33385	5.5879937	2.7235795	4.627604
R5	5,185.74788	4,310.48125	2,664.10185	741.600453	1,137.356526	4,128.23543	269.3591946	47.0297806	356.544272
R6	3,815.45001	6,110.81070	109.82904	1,088.258967	2,878.267795	1,089.33402	520.7049903	28.9899637	74.868402
R7	3,016.76594	3,526.82945	3,007.36873	1,117.861651	4,621.937401	2,886.25820	377.3374861	6.7351814	303.818201
R8	145.37072	284.10230	159.69386	241.381441	35.769227	56.20750	0.7568315	16.3369970	7.700879
R9	1,419.19992	1,928.01908	1,375.04930	3,038.103962	3,956.565316	1,307.53675	294.3925341	72.2655001	314.604309
R10	2,298.72797	3,492.93164	590.41922	3,657.624487	2,389.836823	738.71336	115.7568626	173.9813879	232.787230
R11	2,065.73771	1,018.78684	1,342.48671	1,039.013719	1,671.727623	225.58083	449.2580179	318.9574985	437.525454
R12	7,991.80249	3,686.02952	2,572.10107	6,071.100910	778.607363	1,824.11003	401.5136198	215.9213996	467.654636
R13	4,846.51975	927.54673	3,394.80423	3,148.732569	2,654.500134	3,251.29013	617.5150600	51.7758789	211.730849
R14	1,324.17396	35.07971	446.00009	1,687.130337	338.094351	668.12601	146.6891357	88.9797857	161.227617
R15	30.06423	200.86667	668.40309	1,468.053013	1,049.326003	730.00114	27.8136115	73.1766772	86.445370
R16	2,936.93481	3,797.28938	1,883.23930	2,944.669392	3,234.262561	76.55964	295.6086673	409.0097845	495.133220
R17	7,207.81534	238.23564	351.04323	3,144.350848	3,706.218314	1,491.49425	252.5133816	246.4287582	53.044319
R18	4,622.12194	3,030.54747	2,073.13716	5,882.400548	2,521.409500	353.29743	337.6009217	44.6389951	11.264079
R19	2,874.37305	1,349.09082	3,371.77005	3,299.943150	6,575.235044	903.04392	72.7949400	99.6538830	493.699851

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R20	6,721.11203	3,375.14871	3,347.35632	2,766.887456	4,235.358024	1,212.53459	155.7542298	45.1498178	226.928041
R21	2,061.51173	6,767.99359	1,039.47221	3,360.756794	6,844.697436	2,396.02092	68.3281156	103.8329241	434.413489
R22	4,099.66107	2,346.57908	4,180.56943	5,554.741109	1,500.979625	1,057.13371	244.6450922	3.5064470	570.339514
R23	4.66015	1,946.86832	650.69257	532.566511	1,098.402630	11.27996	145.7324467	47.4850753	168.308541
R24	240.28061	127.47131	69.46127	236.205157	150.229150	102.99883	11.1058005	3.8733706	5.290672
R25	43.37919	41.26992	64.68376	156.409458	62.878608	78.25407	9.7747685	9.1474036	2.885512
R26	134.39666	288.52252	135.63018	43.267034	126.101283	150.10667	26.3169370	0.1698443	31.736531
R27	6,204.85862	1,056.27571	1,390.21600	5,358.797740	4,632.647048	3,611.54528	358.6185664	431.4875803	413.836572
R28	3,329.77822	1,054.27680	834.97651	3,388.795493	5,716.070978	1,370.53208	505.8642672	149.6763085	866.967095
R29	3,108.96217	2,516.96687	1,364.50604	5,055.715124	2,089.200051	1,799.69381	374.8917918	296.4173840	664.115344
R30	128.88633	7,566.01913	1,865.21091	148.351505	1,543.475141	1,207.04907	263.2078023	241.1252610	75.009805
R31	1,249.17144	1,233.79532	463.96765	459.738101	761.424688	652.08528	83.2482202	18.9379032	29.829928
R32	1,247.66434	1,472.32781	797.77160	47.883882	795.085139	84.88587	13.2862023	72.5508131	151.398142
R33	722.64354	4,446.62783	962.19985	2,105.765679	630.142189	168.49421	259.9381976	19.8235783	3.629074
R34	1,021.45497	1,161.67359	1,205.25864	3,029.065899	1,464.327053	335.46053	437.9780529	63.9081899	404.589843
R35	1,924.42195	1,646.36090	1,228.64923	1,931.560403	1,874.013368	237.36168	246.2336369	91.3801912	116.975259
R36	662.51950	2,960.73661	332.24911	2,679.744265	1,980.573975	1,033.48489	128.1001285	125.5031464	24.589884
R37	1,428.17674	2,141.27663	82.53213	3,020.265933	727.591970	1,711.15381	335.6236118	216.6294444	164.319467
R38	1,043.59162	2,066.52535	2,282.63647	3,228.388680	6,983.847565	2,369.01696	611.3654751	72.0231907	100.819683
R39	318.33320	1,630.12033	963.33616	165.854222	6,150.203802	1,059.84055	401.0837875	75.0379060	152.231260
R40	625.38619	509.57462	105.80564	1.317539	611.886584	16.54102	22.7422811	13.9302001	10.631895
R41	644.21077	848.45493	507.29409	1,039.217605	1,054.155379	403.74819	101.2077552	11.0276404	91.864062
R42	246.16166	242.28754	43.19038	206.846466	9.918369	50.91034	8.5897562	13.5508574	33.594153
R43	1,435.67934	1,134.24252	920.24911	674.643500	271.625478	35.84172	105.4580308	103.7778216	63.059631
R44	851.33038	702.12079	468.39390	90.815844	1,494.589118	763.04070	9.1503065	28.8028121	50.433101
R45	185.09901	1,213.63586	608.18798	1,163.901880	1,083.594258	238.57977	101.8360666	33.4600729	62.726569
R46	274.42179	483.17395	23.82443	184.764540	652.311298	160.84901	29.4901141	16.5500380	63.263658
R47	88.03043	773.93979	344.23129	12.953946	376.627919	124.89996	36.5214350	18.7339843	33.702026
R48	234.26357	132.99715	185.59525	174.500059	103.910992	66.38857	33.8696016	12.4132225	22.897872
R49	197.61258	301.56545	206.17491	65.057091	49.086152	84.06705	2.7925650	1.8399199	35.360374
R50	261.56931	135.90490	25.12996	242.904569	119.227185	51.14907	7.9944743	16.1084369	15.919633
R51	12.40338	212.71476	147.24174	226.933194	43.459506	194.59542	19.5167727	8.1793230	21.843635
R52	686.95196	401.91071	1,268.01109	899.262177	878.339356	429.32584	35.6608708	39.2671688	12.085999

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R53	1,085.73873	1,262.22939	775.70403	457.159717	922.106957	82.53886	87.7714545	23.8773137	43.411027
R54	20.00602	477.84224	45.34719	92.417338	82.208260	157.35046	27.3453942	6.2440125	17.727380
R55	196.86522	346.63843	150.33419	107.124863	14.840232	122.82265	3.7302201	16.0604620	17.968568
R56	80.87476	250.46189	241.30910	154.768079	122.931854	49.34263	29.3614694	6.1398602	30.870757
R57	970.05708	578.02087	316.59972	1,242.115803	774.976307	858.01495	21.1339819	10.5977647	111.697622
R58	4,573.44247	1,926.78684	2,477.14353	2,758.576780	827.365446	3,433.90230	76.2408379	330.2752942	143.426812
R59	188.81594	1,090.86180	518.01393	1,276.294889	925.547094	629.27412	4.9079501	70.7844223	15.508818
R60	80.56442	200.72606	25.40637	432.476566	25.722843	101.51748	42.1356464	0.8690266	32.553056

11.3 Chebyshev Goal Programming Solution

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R2	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R3	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R4	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R5	0.000	0.0000	1,234.6787	0.0000	0.0000	0.000	0	0	0	0.0000
R6	0.000	0.0000	7,958.0429	0.0000	12,041.9571	0.000	0	0	0	0.0000
R7	0.000	0.0000	8,359.9383	0.0000	0.0000	1,890.062	0	0	9,750	0.0000
R8	0.000	0.0000	0.0000	0.0000	0.0000	1,000.000	0	0	0	0.0000
R9	0.000	102.3406	0.0000	14,897.6594	0.0000	0.000	0	0	0	0.0000
R10	0.000	0.0000	0.0000	15,000.0000	0.0000	0.000	0	0	0	0.0000
R11	40,000.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R12	32,528.357	38,998.6023	0.0000	0.0000	8,473.0411	0.000	0	0	0	0.0000
R13	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	1,100	0	0.0000
R14	0.000	0.0000	0.0000	0.0000	0.0000	0.000	5,000	0	0	0.0000
R15	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R16	0.000	1,231.6725	0.0000	0.0000	0.0000	0.000	0	0	0	18,768.3275
R17	0.000	0.0000	0.0000	0.0000	0.0000	3,082.454	3,000	3,900	0	0.0000
R18	3,328.753	0.0000	0.0000	0.0000	16,671.2474	0.000	0	0	0	0.0000
R19	0.000	0.0000	0.0000	0.0000	20,000.0000	0.000	0	0	0	0.0000

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	0.000	0.0000	0.0000	40,000.0000	0.0000	0.000	0	0	0	0.0000
R21	0.000	18,417.3750	18,134.9499	3,447.6751	0.0000	0.000	0	0	0	0.0000
R22	0.000	0.0000	0.0000	7,615.9712	7,509.4432	0.000	0	0	0	0.0000
R23	0.000	0.0000	0.0000	5,000.0000	0.0000	0.000	0	0	0	0.0000
R24	0.000	0.0000	0.0000	0.0000	516.0415	0.000	0	0	0	0.0000
R25	0.000	0.0000	0.0000	0.0000	500.0000	0.000	0	0	0	0.0000
R26	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R27	0.000	0.0000	21,460.7884	0.0000	0.0000	14,027.485	0	0	0	0.0000
R28	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R29	0.000	0.0000	0.0000	0.0000	0.0000	25,000.000	0	0	0	0.0000
R30	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R31	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R32	0.000	0.0000	0.0000	0.0000	0.0000	5,000.000	0	0	0	0.0000
R33	0.000	0.0000	0.0000	0.0000	0.0000	0.000	2,000	0	0	0.0000
R34	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R35	0.000	0.0000	0.0000	0.0000	624.2942	0.000	0	0	0	0.0000
R36	0.000	4,571.0624	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R37	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R38	14,133.729	0.0000	0.0000	1,902.9377	0.0000	0.000	0	0	0	0.0000
R39	0.000	0.0000	0.0000	0.0000	15,000.0000	0.000	0	0	0	0.0000
R40	0.000	2,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R41	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R42	0.000	0.0000	339.4425	0.0000	0.0000	0.000	0	0	250	410.5575
R43	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R44	0.000	0.0000	0.0000	522.9126	4,477.0874	0.000	0	0	0	0.0000
R45	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R46	0.000	0.0000	0.0000	2,000.0000	0.0000	0.000	0	0	0	0.0000
R47	0.000	0.0000	0.0000	0.0000	2,000.0000	0.000	0	0	0	0.0000
R48	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R49	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R50	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R51	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	821.1150
R52	0.000	2,487.8407	2,512.1593	0.0000	0.0000	0.000	0	0	0	0.0000

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R54	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R55	0.000	529.7838	0.0000	470.2162	0.0000	0.000	0	0	0	0.0000
R56	0.000	754.9784	0.0000	245.0216	0.0000	0.000	0	0	0	0.0000
R57	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R58	10,009.162	28,906.3443	0.0000	0.0000	1,084.4942	0.000	0	0	0	0.0000
R59	0.000	0.0000	0.0000	3,897.6061	1,102.3939	0.000	0	0	0	0.0000
R60	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

11.4 Genetic Algorithm Solution

Table 11: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R1	448.35001	772.86582	867.83084	343.74477	504.70205	140.23870	443.97761	393.57429	520.6661
R2	110.28589	110.75958	72.85369	93.54542	108.53056	75.08336	134.30017	128.50834	74.4775
R3	2,096.42916	2,616.18763	2,538.03326	3,450.19196	4,461.61787	2,002.49898	1,798.25037	3,886.04708	1,848.2859
R4	92.12356	100.48671	91.53613	92.94451	99.38798	119.33927	109.18655	105.52768	75.5357
R5	3,676.87641	3,254.98444	2,339.28835	2,304.29715	2,326.39073	2,479.68912	2,593.66991	2,430.40869	1,718.1560
R6	2,383.98534	2,009.76340	2,759.43867	2,712.30891	1,939.31570	1,628.06910	2,061.37603	1,900.74593	1,487.2762
R7	1,749.65412	2,375.81209	2,405.27236	2,335.30388	2,367.43584	897.66427	1,996.51148	2,216.12527	2,322.6566
R8	109.57926	84.73293	109.14557	51.60957	97.36922	113.63205	116.77752	156.01181	44.0470
R9	2,207.30909	1,502.77324	936.64547	2,031.03342	1,914.95907	1,624.57722	1,108.28838	983.82738	1,704.7171
R10	2,324.51408	1,297.09147	993.86277	1,307.73704	1,490.95847	771.11705	2,336.64616	1,954.67313	1,864.9204
R11	6,084.58893	3,345.66592	1,876.02961	6,791.97188	6,758.48744	3,718.98104	2,412.66521	1,145.74685	3,945.9869
R12	11,978.51086	10,547.65440	6,026.43329	10,946.08778	11,429.90598	8,759.04800	6,484.39482	6,507.45856	3,955.8757
R13	6,764.57131	10,916.70264	4,569.00849	8,617.68172	10,072.73711	4,600.12441	4,412.72047	3,413.85301	3,451.9617
R14	390.00432	642.48166	438.25832	538.27547	567.91483	260.77692	437.74305	526.03353	666.5355
R15	765.36515	562.10522	471.84878	516.48683	548.06548	475.31025	625.56541	464.57529	467.9818
R16	2,156.92782	2,000.32786	848.52109	2,861.59720	1,511.96735	2,614.01029	2,066.48824	2,836.84607	1,571.9802
R17	1,525.90878	3,364.28513	1,833.36261	2,796.51476	1,812.15789	1,309.73877	2,703.48766	2,640.10728	835.0060
R18	2,625.62358	1,603.49416	2,029.57770	2,137.45856	2,210.92644	1,656.83306	2,656.88810	1,658.28225	1,524.0198
R19	2,620.39834	1,431.13554	2,334.30367	2,252.31027	1,648.61464	2,138.56312	2,117.59575	1,542.30244	2,176.6850

Table 11: GA Solution: Amount of Each Material in Each Blend
(60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R20	5,665.65836	6,146.01618	3,449.61760	5,367.99269	5,162.09933	3,991.84596	3,393.30141	1,958.64247	1,903.3325
R21	3,972.74713	5,294.53296	4,655.16361	5,320.74177	3,981.96108	3,044.28881	3,820.23197	3,300.88683	3,477.2461
R22	2,200.89264	2,243.18468	1,960.39753	1,949.31591	2,503.90617	1,428.26928	1,250.61388	2,009.54900	1,913.2353
R23	655.62679	450.57101	621.85903	534.73936	439.74353	436.41296	503.70687	540.30035	398.6208
R24	142.47162	80.52634	146.20803	97.31180	80.23526	71.09143	106.20290	83.82706	124.0215
R25	74.26734	27.15856	54.57787	64.97698	53.14591	54.02690	53.67103	38.95934	48.3267
R26	69.39105	162.41643	103.92169	45.57768	146.18697	125.99327	125.49938	66.29209	88.2100
R27	3,849.36995	5,763.74465	3,173.93522	4,342.18318	3,818.27167	2,866.41023	2,362.82221	4,910.10239	4,094.8456
R28	4,002.38842	3,642.26189	1,542.32323	6,443.39473	5,363.80601	5,195.96306	4,049.78720	5,396.99247	3,208.5291
R29	3,346.79911	3,552.01464	2,949.67547	2,160.47765	1,969.89083	2,989.31223	1,863.86800	1,984.03336	1,384.0182
R30	4,141.68021	5,020.12870	2,500.29515	2,929.93068	3,347.09064	2,928.93527	2,429.59813	1,211.76221	2,704.4799
R31	768.17165	555.90937	490.31962	358.46036	285.76322	416.45260	563.15205	386.65758	555.0965
R32	573.48070	649.39343	481.31822	367.67503	587.23601	399.19735	544.78414	377.69823	513.6625
R33	1,042.47557	946.10141	949.04716	1,133.44963	1,259.55768	823.70430	1,351.99411	901.76104	955.1021
R34	1,680.89968	935.89677	739.12824	1,160.78891	1,190.78229	1,051.48491	666.77026	960.57746	912.8731
R35	1,408.44291	1,376.03229	1,075.80811	788.36070	1,287.12827	920.70236	1,095.17728	906.64945	859.8958
R36	1,696.20095	976.03437	1,006.55260	1,009.65040	1,085.02931	1,202.77414	1,087.52342	382.76792	1,211.2682
R37	1,659.45386	730.53074	1,015.65786	782.93488	1,573.90627	924.39162	721.32054	1,029.68196	886.3745
R38	2,478.70861	2,459.96787	2,247.43899	1,231.05085	2,401.26207	1,719.77197	3,008.77468	2,558.21540	738.0624
R39	2,104.69156	1,073.38656	729.11737	1,644.09814	1,662.72503	2,185.95456	1,705.70488	1,571.49990	1,130.1235
R40	208.93822	175.79913	191.79136	183.28671	240.42600	286.66745	198.11184	145.67781	249.4650
R41	540.00164	436.68457	522.78089	584.90727	538.84740	363.63229	464.69022	765.21351	395.4139
R42	94.40846	72.23645	100.10022	129.04785	133.46827	92.96418	99.03789	101.88783	116.7086
R43	872.42663	364.09106	682.11946	331.60290	465.83095	581.44698	227.20862	373.04913	761.9642
R44	427.55962	471.37001	639.28560	584.46847	635.49472	518.27316	393.83700	573.15345	296.8422
R45	435.90229	447.46529	687.63459	375.21435	301.13245	787.95023	647.17927	505.63007	283.0446
R46	214.12769	275.87696	195.39732	263.53855	151.52243	156.12391	119.44581	301.82887	166.1889
R47	145.98001	200.69912	229.31556	254.88111	268.82637	164.02864	173.08156	154.10907	195.2515
R48	115.87667	70.35954	56.43237	119.63223	128.88955	145.68601	113.25108	62.64156	144.9243
R49	131.16261	109.99852	64.64956	78.89853	84.05558	87.45014	94.15147	137.85499	112.0500
R50	114.86395	99.61252	144.46557	44.81564	136.77961	80.86356	109.04205	126.43508	76.7461
R51	113.19524	69.38835	125.18760	96.66824	111.95720	73.36157	49.77324	108.91187	139.6779
R52	318.79773	758.49170	207.78324	288.43606	411.40005	478.40375	575.44545	763.76922	484.4155

Table 11: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B
R53	563.57254	374.07942	659.44398	350.41512	454.33453	521.86593	662.40371	430.09776	648.0092
R54	120.61638	131.85627	117.20664	110.99422	106.12869	62.35613	112.43381	95.30714	88.9120
R55	117.27427	70.73409	57.07972	124.44387	111.66220	126.60065	122.07880	90.55349	81.1569
R56	111.49985	103.69880	65.69137	141.43770	77.46980	129.69942	126.54101	72.29287	96.7238
R57	557.52318	569.41420	669.49473	690.95985	687.66666	571.95538	459.38122	188.26264	202.1189
R58	2,489.45155	3,974.42059	4,619.80689	4,531.42379	4,692.21778	4,615.06363	3,738.26124	4,583.66026	2,697.1307
R59	534.06520	477.61951	466.82392	680.00785	699.34684	423.07886	436.45573	508.13390	455.6882
R60	127.93214	120.98520	87.89327	120.70725	68.68555	91.38265	78.70079	69.44695	165.7146

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